The mass of an asymptotically flat manifold from an extrinsic point of view

Girao, Frederico, de Sousa, Alexandre

doi:10.3842/SIGMA.2025.018; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned journal source file sources/sigma-2025-018-mass-from-an-extrinsic-point-of-view/pinned-original-source.tex (Mass from an Extrinsic Point of View); no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: All delivered mathematical text is the author own, copied verbatim from the pinned journal source. Statement source lines 480-498 with the author own proof at 500-524 (label 'main theorem'). Printed number 3.1 from the source walk; verified against the host aux. Pre-dispatch self-certification: DOI-wide key scan found no other source carrying this face.

Editorial note. The label main theorem names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own Theorem environment in the article source.

Editorial note. Journal-unit assembly: card-local preamble (author theorem environments and macros verbatim), article class, bibliography backend none; any literature citation carried by the delivered span is disclosed in the notes of this card.

Editorial note. Author names were transliterated to ASCII for the card face by stripping diacritics (1 name(s) affected); the SOURCE.json metadata keeps the original accented form.

Editorial scope. the counted statement and its own proof environment only

Allowed standard prerequisites (editorial).

- The ADM mass and the Gauss-Bonnet-Chern mass of an asymptotically f.... Supplies background that the delivered proof uses. The entry is declared as a dependency; it is not re-proved on the card, and the load-bearing steps of the counted statement are carried out inside the delivered span.

Source reference. Original reference 2.8: sources/sigma-2025-018-mass-from-an-extrinsic-point-of-view/pinned-original-source.tex, source lines 289–289. Navigation pointer invoked at line(s) 505: the counted text refers to this environment

of the same journal unit, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it (the section and the place of the statement inside that section, derived from the shared section-scoped theorem counter of the source); the cited item is an allowed input and none of it is reproduced inside the span.

Source reference. Original reference 3.4: sources/sigma-2025-018-mass-from-an-extrinsic-point-of-view/pinned-original-source.tex, source lines 415–415. Navigation pointer invoked at line(s) 517: the counted text refers to this environment of the same journal unit, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it (the section and the place of the statement inside that section, derived from the shared section-scoped theorem counter of the source); the cited item is an allowed input and none of it is reproduced inside the span.

Source reference. Original reference 3.2: sources/sigma-2025-018-mass-from-an-extrinsic-point-of-view/pinned-original-source.tex, source lines 381–381. Navigation pointer invoked at line(s) 523: the counted text refers to this environment of the same journal unit, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it (the section and the place of the statement inside that section, derived from the shared section-scoped theorem counter of the source); the cited item is an allowed input and none of it is reproduced inside the span.

Source reference. Original reference 2.8: sources/sigma-2025-018-mass-from-an-extrinsic-point-of-view/pinned-original-source.tex, source lines 289–289. Navigation pointer invoked at line(s) 505: the counted text refers to this environment of the same journal unit, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it (the section and the place of the statement inside that section, derived from the shared section-scoped theorem counter of the source); the cited item is an allowed input and none of it is reproduced inside the span.

Source reference. Original reference 2.6: sources/sigma-2025-018-mass-from-an-extrinsic-point-of-view/pinned-original-source.tex, source lines 203–203. Navigation pointer invoked at line(s) 523: the counted text refers to this environment of the same journal unit, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it (the section and the place of the statement inside that section, derived from the shared section-scoped theorem counter of the source); the cited item is an allowed input and none of it is reproduced inside the span.

Source reference. Original reference 3.1: sources/sigma-2025-018-mass-from-an-extrinsic-point-of-view/pinned-original-source.tex, source lines 366–366. Navigation pointer invoked at line(s) 523: the counted text refers to this environment of the same journal unit, which is not retained in this card, so the reference is declared as a pointer carrying the number the article itself prints for it (the section and the place of the statement inside that section, derived from the shared section-scoped theorem counter of the source); the cited item is an allowed input and none of it is reproduced inside the span.

Theorem 3.6. *Let n and q be integers such that $n \geq 3$ and $0 < q < n/2$. If $\psi\colon M^n \hookrightarrow \left(\mathbb{R}^d, \bar{\delta}\right)$, $d > n$, is an asymptotically Euclidean immersion of order*

$\tau > \tau_q$ such that $L_{(q)}$, the q-th Gauss-Bonnet curvature of the induced metric $\psi^*\bar{\delta}$, is integrable, then the functions S_{2q} and $\langle S_{2q+1}, \bar{Z}\rangle$ are integrable and the q-th Gauss–Bonnet–Chern mass of $\big(M, \psi^*\bar{\delta}\big)$ satisfies the following integral identity:

$$\mathrm{m}_q\big(M, \psi^*\bar{\delta}\big) = a(n,q)\bigg[(n-2q)\int_M S_{2q}\,\mathrm{d}M + (2q+1)\int_M \langle S_{2q+1}, \bar{Z}\rangle\,\mathrm{d}M\bigg],$$

where $\mathrm{d}M$ is the Riemannian measure on $\big(M, \psi^*\bar{\delta}\big)$ and

$$a(n,q) = \frac{(2n)!(n-2q-1)!}{2^q(n-1)!\,\omega_{n-1}}.$$

Proof. Let E be one of the ends of M. Consider the function $f\colon M \to \mathbb{R}$, with

$$f = \frac{\psi^*\bar{\rho}^2}{2}.$$

Since $\psi\colon M^n \looparrowright \mathbb{R}^d$ is an asymptotically Euclidean immersion of order $\tau > \tau_q$, the function f satisfies (2.8), and hence, by Corollary 2.8,

$$\mathrm{m}_q\big(E, \psi^*\bar{\delta}\big) = -b(n,q)\lim_{\rho\to\infty}\int_{S_\rho} G_{(q)}(\nabla f, \nu)\,\mathrm{d}S_\rho, \tag{3.1}$$

where the geometric quantities in the integral are computed with respect the induced metric $\psi^*\bar{\delta}$.

Note that

$$\nabla f = \psi^*\bar{Z}^\top.$$

Thus, applying equation (3.1) to all the ends of M together with the divergence theorem and identity (3.4), we find

$$\mathrm{m}_q\big(M, \psi^*\bar{\delta}\big) = a(n,q)\int_M \big[(n-2q)S_{2q} + (2q+1)\big\langle S_{2q+1}, \bar{Z}\big\rangle\big]\mathrm{d}M. \tag{3.2}$$

It remains to show that both S_{2q} and $\langle S_{2q+1}, \bar{Z}\rangle$ are integrable, so the right hand side of (3.2) can be broken in two. The integrability of S_{2q} follows from (2.6), (3.1), (3.2) and the integrability of $L_{(q)}$. Once we know that $S_{(2q)}$ is integrable, the integrability of $\langle S_{2q+1}, \bar{Z}\rangle$ follows from the fact that the left hand side of (3.2) is finite. □